

UTM TENSILE REPORT — UTM_Test_20260822_181957

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A_0) · anchor self-calibration

40.5 MPa

UTS

28.7 MPa

Yield σ_y

2.12 GPa

Modulus E

3.8 %

Failure ϵ_f

1405 kJ/m³

Toughness

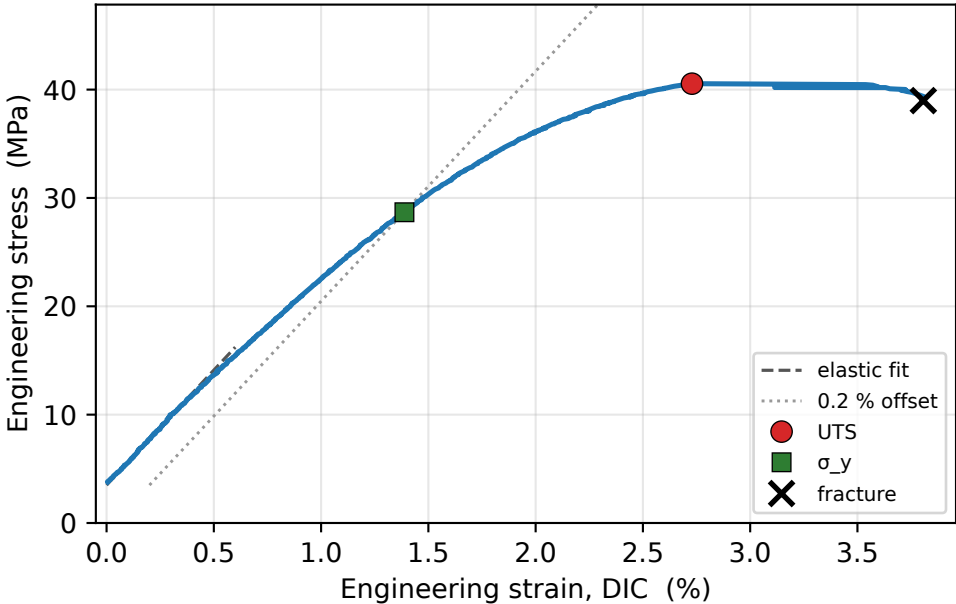
301 N

Preload anchor

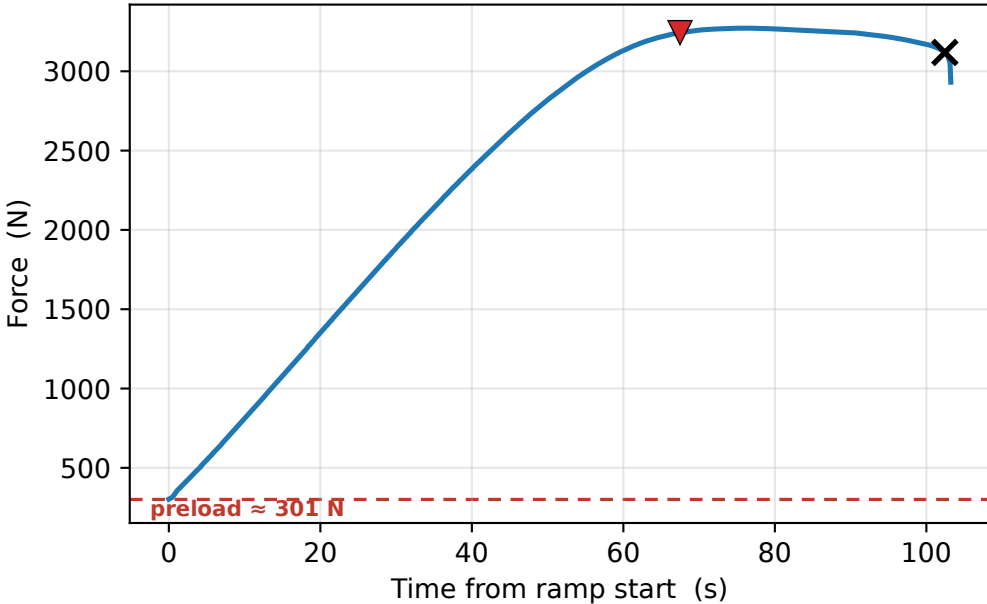
TEST SETTINGS

Specimen	UTM_Test_20260822_181957
DIC mode	Black
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 325.26
Comment	—
Test date	2026-08-22 18:18:04
Duration	121.8 s

Stress-strain



Load vs time

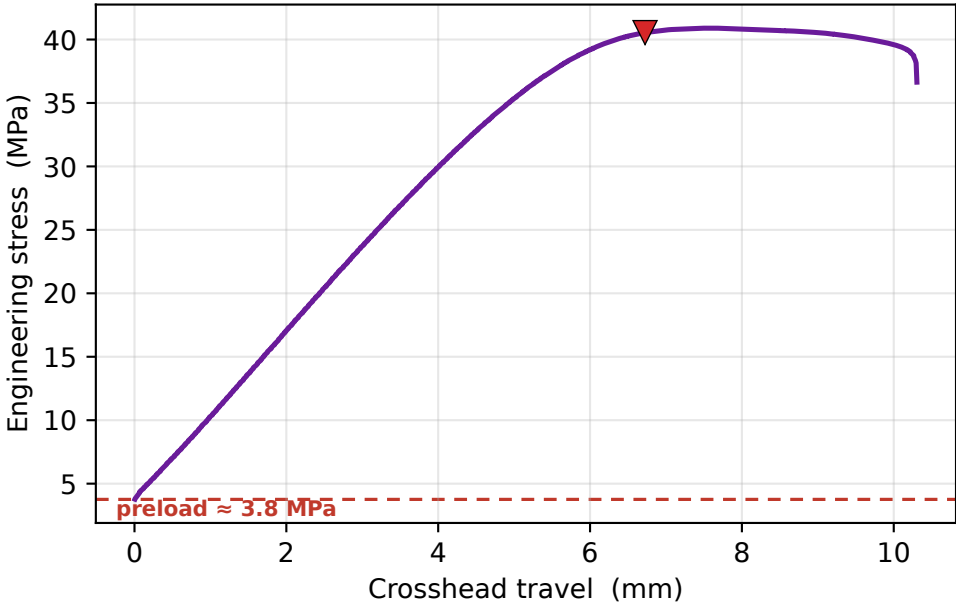


VALIDATION vs REFERENCES

prop	meas	Chacón	E-PLA k
UTS	40.5	32-60 PASS	×1.43
σ_y	28.7	30-50 low	×2.02
E	2.1	3-5.5 low	×1.35
ϵ_f	3.8 %	(range 3-7 %)	×2.10

⚠ strength outside journal range

Stress vs crosshead



DIC strain vs crosshead — the compliance gap

